# Dummy-Trajectory Synthesis: A Privacy-Preserving Approach for Semantic Trajectory Data in IoT-Based LBSN

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***Abstract*—Trajectory data analysis is crucial in various applications but presents significant privacy risks, as location data can reveal sensitive information. Existing privacy protection methods, such as spatiotemporal *K*-anonymity and *L*-diversity, are vulnerable to semantic inference attacks, where public data is exploited to re-identify users. To address these challenges, we propose dummy-trajectory synthesis (DTS), an efficient privacy protection scheme for location-based social networks (LBSNs). DTS enhances privacy by leveraging users' frequent behavioral sequences to generate synthetic dummy trajectories. Unlike traditional approaches, DTS considers both geographic and semantic data by segmenting historical trajectories into time periods using the OPTICS clustering algorithm. This enables the identification of regions with specific semantic attributes and the mining of semantic trajectory sequences. DTS optimizes dummy trajectory generation by combining Euclidean distances and semantic similarity, ranking historical points and establishing transition relationships. Experimental results show that DTS significantly improves privacy protection and performance compared to existing methods, without compromising service quality. DTS offers a robust solution for protecting trajectory data privacy in LBSNs against transition probability attack for joint time periods.**

***Index Terms*—Frequent behavioral sequence, privacy protection, semantic trajectory sequence, trajectory pattern.**

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## I. Introduction

In THE backdrop of rapid advancements in wireless networks, mobile technologies, and the Internet of Things (IoT) [1], [2], [3], coupled with the pervasive adoption of smart devices, location-based social network (LBSN)—platforms that combine geographic location data with social networking features, such as location-based check-ins and recommendations—have become ubiquitous in our daily lives. These LBSN applications find multifaceted utility, ranging from car navigation and mobile phone positioning to personalized food and scenic spot recommendations [4]. However, while users enjoy the convenience provided by LBSNs, their personal privacy (such as identity and location information) is continually exposed to significant security threats [5], [6], [7]. To access the social networking services provided by LBSN applications, users must transmit their spatiotemporal data to the remote servers of these applications. Attackers can exploit this private data to analyze users' behavioral patterns, interest preferences, and the potential relationships between different locations [8], [9], [10]. Furthermore, additional personal sensitive data, such as health status, social relationships, professional details, and residential addresses, may also be maliciously utilized by attackers [11]. Therefore, it is crucial to address the challenge of maintaining user privacy, particularly location information, while utilizing LBSN services.

In recent years, scholars have proposed a variety of location-based privacy protection schemes, with the *K*-anonymity model being the most extensively discussed and adopted, alongside *L*-diversity privacy protection schemes [12]. The main content of these two schemes is to anonymize the exact location of the mobile user by blurring the exact location to the cloaking region (CR) [13]. In the *K*-anonymity model, exact user locations are grouped with at least $K-1$ other users to enhance user protection. *L*-diversity requires diverse sensitive attributes within each equivalence class, which is a group of records that share the same values for certain identifying attributes. This reduces the probability of attackers identifying sensitive information to $1/L$, thus improving user privacy.

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While these methods have been robust, recent advancements in semantic analysis and data mining reveal limitations.

These limitations lead to significant direct consequences. For instance, attackers [13] can perform semantic inference attacks by combining anonymized data with publicly available information, thereby reidentifying users or deducing sensitive information about them. These attacks often involve filtering or identifying location points based on semantic context, which can reduce the number of fake locations to fewer than $K-1$ in $K$-anonymity, or reduce the diversity within equivalence classes to fewer than $L$ distinct sensitive attributes in $L$-diversity. This undermines the privacy guarantees that $K$-anonymity and $L$-diversity aim to provide. Furthermore, current approaches predominantly involve the direct or simplistic processing of raw datasets. When dealing with massive datasets like Gowalla [14] and Foursquare [15], this method significantly diminishes efficiency.

Given the evolving landscape of privacy threats, especially in semantic inference, it is crucial to develop advanced privacy-preserving [32] methods that address these limitations. Recognizing these challenges, we propose a novel approach that considers not only the spatial and temporal factors but also the semantic context of location data to enhance privacy protection. Our approach aims to improve the generation efficiency and query service quality of dummy trajectories in the context of privacy protection. By leveraging user frequent behavior patterns, we enhance the preservation of user privacy. We not only consider the basic geographic information of user historical trajectories but also the semantic information contained in the location, such as the semantic characteristic that categorizes breakfast and fast food establishments within the domain of dining establishments.

We recognize that user behavior patterns vary across different time periods, motivating our approach to segment historical trajectories into distinct time periods. To achieve this, we employ the OPTICS clustering algorithm to identify density points and cluster trajectories, thereby deriving candidate sets of regions, each characterized by its semantic attributes. Subsequently, we mine semantic trajectory sequences to extract frequent user behavior patterns.

To address the efficiency challenge, we have adopted an optimization strategy that determines the optimal time intervals and regions based on user trajectories. Specifically, this optimization process encompasses the calculation of Euclidean distances between historical and anonymous location points. Furthermore, we leverage a semantic similarity algorithm to quantify the semantic resemblance between these points. Previous research has predominantly focused on the utilization of geographical distance while overlooking semantic considerations [17]. Addressing this gap, we rank historical location points by combining geographical distance and semantic similarity, resulting in a candidate location set. Subsequently, we determine transition relationships between regions and location points based on semantics and frequent behavior patterns. Finally, we constitute the candidate trajectory set based on the mined frequent behavior patterns and the established transition relationships between regions and location points. Our proposed method aims to address the requirements of $L$-semantic diversity and location physical discreteness while enhancing the efficiency of generating anonymous trajectory sets. The main contributions of this article are summarized as follows.

1) We extract frequent user behavior patterns from their historical semantic trajectory datasets. These patterns are then integrated into our model for determining transitions between candidate regions and relationships among candidate location points. This approach enhances the efficiency of generating dummy trajectories by considering both spatial and semantic aspects, addressing the need for effective privacy protection.
2) We establish transition relationships among candidate regions for each time period and among candidate locations. To improve matching efficiency, we introduce a dynamic matching algorithm that accommodates frequent behavior patterns of varying lengths. Empirical results confirm the effectiveness of our proposed method, demonstrating reductions in execution time and improvements in overall execution efficiency.
3) Our approach specifically addresses a range of privacy threats, with a focus on safeguarding against transition probability attack for joint time periods. By leveraging the semantic context of location data, our method enhances the robustness of data protection measures without compromising the quality of user services. In fact, our approach shows notable improvements in service quality while ensuring strong privacy protection.

The subsequent discourse of this work unfolds in a structured manner. The rest of this work is organized as follows. In Section II, we summarize related works. Section III-A presents the symbology employed and key definitions. Section III-B introduces the attack method. We introduce the proposed model in Section III. Finally, Section IV discusses the empirical results with a specific analysis, and Section V concludes this work and discusses future work.

## II. Related Work

In the context of LBSN, safeguarding user privacy has been a paramount concern in the era of big data. To address this complex issue, researchers have proposed various strategies and methods to enhance privacy protection.

*K-Anonymity*: $K$-anonymity is a widely adopted approach for privacy preservation, aimed at processing published anonymous datasets. Its core principle is to ensure that each published data point remains indistinguishable from at least $K-1$ other data points within the same anonymized set [18], [19]. Luo et al. [20] considered the user's trust level when constructing the $K$-anonymous stealth area and protected the user's location information privacy by effectively preventing malicious users. Majeed and Hwang [21] proposed a hybrid anonymization scheme combining DP and $K$-anonymity, which optimizes the balance between privacy protection and data utility by dividing privacy violation and non privacy violation zones, improving data practicality and privacy protection effectiveness. Hong et al. [22] combined $K$-anonymity with game theory based dynamic strategies to

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optimize privacy protection by utilizing the flexibility of users in service requirements, and achieved good privacy protection effects even when the $k$ value was small. However, certain implementations of $K$-anonymity, such as those based on region of interest partition and virtual location generation, raise concerns related to data authenticity and susceptibility to attacks [23].

*Semantic Similarity Approach*: Existing research diverges from traditional $K$-anonymity by placing a strong emphasis on harnessing semantic similarity among geographical location points. This innovative approach integrates the multicenter clustering algorithm, culminating in a method that surpasses the mere attainment of $K$-anonymity. For example, Qiu et al. [24] proposed a method that uses semantic similarity to select candidate locations from a user's historical track set. This allows for more nuanced generalization, where locations with similar functions are grouped together, further enhancing privacy. For example, both Mexican restaurants and Western restaurants can be categorized as "restaurants," thereby generalizing the specific type of cuisine while preserving the core semantic meaning. This method involves data concealment, substitution, and generalization to protect user privacy. However, these methods may not fully consider the user's mobile behavior and habits, potentially leading to the inadvertent disclosure of privacy data.

*Strengthening Location Privacy Through Trajectory Transition Characteristics*: Li et al. [25] have introduced a virtual user mechanism to generate virtual users who may be located in different cities, and proposed a virtual trajectory generation algorithm to ensure that the generated trajectories are consistent with real user behavior, thereby enhancing anonymity protection. Shen et al. [26] introduced a location privacy protection mechanism that emphasizes user trajectory transition characteristics within a designated time period to enhance privacy safeguards. Siddiqie et al. [27] proposed the AGDG method, which introduces time-correlated infeasible regions to enhance the final anonymity protection effect. Qiu et al. [11] presented BSPri, a posterior behavioral-semantic privacy-preserving approach aimed at addressing privacy challenges in mobile Internet services. BSPri employs techniques, such as similar semantic clustering, ranking, and synthetic trajectory generation to mitigate potential inferential attacks. Zhang et al. [28] introduced a novel context-based mix strategy for enhancing privacy in VANETs, mitigating tracking risks and reducing information loss during pseudonym changing. However, it is crucial to acknowledge that their approach may have inherent limitations in countering semantic attacks. To address challenges associated with generalization and resistance to inference, recent studies have proposed the integration of causal modeling and generative frameworks. The ITCRL-DGAN model [29] introduces information-theoretic and causal representation learning into dual generative adversarial networks. This approach improves out-of-distribution (OOD) generalization through causal feature separation and counterfactual data augmentation, thereby enhancing privacy protection in smart transportation and industrial settings.

*Leveraging Differential Privacy*: Cai et al. [30] proposed a differential privacy algorithm called DPTD, which constructs a 3-D generalized trajectory dataset, partitions the data based on timestamps, and combines clustering and generalization techniques to achieve privacy protected trajectory data publishing. In addition, the algorithm also introduces a prediction method based on Markov assumption, effectively reducing noise addition and saving privacy budget. Ghane et al. [31] employed differential privacy to process datasets, capture the user's motion trajectory, and synthesize generalized trajectory sets while retaining original spatiotemporal information. Wu et al. [32] proposed TCPP that ensures differential privacy for multiuser trajectory correlations while maintaining data availability and efficiency, addressing privacy risks in location-based services caused by trajectory data sharing. Zhang et al. [33] proposed CTDP based on differential privacy, utilizing trajectory clustering, personalized privacy budget allocation, and random sampling of confusion points to balance privacy protection and data availability. This approach is a significant step in privacy-preserving location-based services.

Meanwhile, recent advances demonstrate that combining differential privacy with structural and distributed models can further enhance security. For instance, the DFGL-LZTA model [34] integrates decentralized federated graph learning with a lightweight zero trust architecture. It employs deep reinforcement learning for policy updating and hierarchical graph attention networks to improve both data confidentiality and system efficiency in distributed LBSN environments. In addition, the RG-GLD framework [35] introduces a lightweight anomaly detection method combining graph neural networks with knowledge distillation. Through graph reconstruction and local–global attention, this model achieves high detection accuracy in dynamic IoT networks while minimizing computational overhead.

*Federated Learning in Privacy-Sensitive Applications*: In the context of the Internet of Medical Things (IoMT), where privacy concerns are amplified by data sensitivity and device heterogeneity, federated learning has emerged as a key enabler of secure, decentralized collaboration. A hierarchical federated learning framework [36] incorporating social context clustering enables efficient participant selection, addressing the accuracy degradation caused by low-quality clients and supporting model aggregation across edge networks. Further, the FDBC-SKS scheme [37] combines federated distillation and blockchain to realize secure knowledge sharing. Its reinforcement-learning-enhanced consensus mechanism addresses fairness, scalability, and load-balancing challenges in IoMT data sharing.

*Personalized Federated Learning in the Metaverse*: To address user heterogeneity and cross-modal data distributions in human-centric Metaverse environments, personalized federated learning (PFL) has gained attention. The PFL-MCL framework [38] integrates model-contrastive learning with hierarchical personalization and multicenter aggregation, allowing adaptive, privacy-preserving training across large-scale 5G and beyond networks. This approach supports efficient communication, personalized user modeling, and secure data processing in immersive applications.

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While the existing methods for synthesizing generalized trajectory sets have been effective to some extent, they suffer from certain limitations that need to be addressed. The prevalent approach involves finding corresponding replacement position points and replacing them on the trajectory one by one, which, unfortunately, results in a significant increase in execution time.

However, our innovative approach not only improves efficiency by segmenting trajectories based on time periods but also incorporates semantic characteristics and the global context of the entire trajectory. By doing so, we reduce execution time and significantly enhance the overall efficiency of trajectory synthesis while considering both semantic and holistic trajectory features. This holistic approach contributes to a more comprehensive and effective solution in the context of LBSN privacy preservation.

## III. Our Proposed Privacy-Preserving DTS

This section presents the dummy-trajectory synthesis (DTS) privacy-preserving protection scheme, which aims to enhance user privacy by generating synthesized trajectories that preserve both spatial and behavioral patterns. To provide a comprehensive understanding of the method, we first introduce our system model and some key concepts. We also present an attack methodology to illustrate the potential privacy risks and motivate the need for our DTS scheme. The proposed approach is composed of three main modules: 1) user modeling; 2) region candidate set construction; and 3) generalized trajectory synthesis. These modules work together to address challenges, such as large data volumes and maintaining behavioral consistency, ensuring robust privacy protection for users in LBSN.

### A. Preliminary

In the context, *semantic trajectory data* refers to the data that captures not only the geographical locations but also the semantic attributes associated with these locations, such as the type of place (e.g., restaurant, park, or office). This enriched data provides deeper insights into user behavior and context. On the other hand, *dummy trajectories* are synthetic trajectories generated to obscure real user movements while preserving the utility of the data for services. These dummy trajectories are designed to protect user privacy by making it difficult for attackers to infer sensitive information from the data.

To provide a comprehensive understanding of our proposed method, we introduce additional basic concepts and notations in Table I, which will be used throughout this article.

### B. Attack Methodology

The attacker infers users' private data by analyzing the historical trajectory sets and published anonymized sets, leveraging background knowledge [23], [39]. So that, we propose an attack method based on this context: the *Transition Probability Attack for Joint Time Periods*.

This attack specifically targets the lack of protection in existing methods, particularly with respect to the semantic angle and, more crucially, the temporal-semantic aspect. As illustrated in Fig. 1, the user's trajectory consists of leaving an apartment at 16:00, arriving at a park at 18:10, dining at a restaurant at 19:00, and eventually heading to a bar. Previous protection methods either ignore semantic transitions, focusing solely on geographical factors, or consider semantic transitions while neglecting the temporal component. For instance, if the anonymized set includes a school, but the user's visit period to the school is not in the evening, an attacker, utilizing such background knowledge, can identify false locations, thereby undermining the protection.

As shown in Algorithm 1, the proposed method calculates the transition probability between each position in the anonymized set and the original position to determine which positions are likely to be recognized. If the transition probability for a given location is below a threshold $\theta$ and the location is present in the candidate set $S_t$, the location is considered identified. The algorithm ultimately outputs the number of identified positions. In our implementation, the threshold $\theta$

TABLE I
Basic Concepts and Symbols

| Symbol | Explanation |
|---|---|
| $\Omega$ | User trajectory dataset. |
| $tra_i$ | User's trajectory. |
| $t^u$ | Real user trajectory. |
| $T$ | Dummy anonymous set. |
| $T'$ | Final anonymous set. |
| $O_s$ | Location semantic set. |
| $tra_k$ | Trajectory in $T'$. |
| $loc_i$ | Location point. |
| $time_i$ | Visit time. |
| $sem_i$ | Location semantic. |

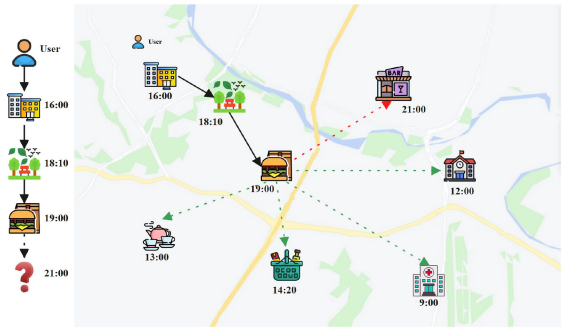



Fig. 1. Trajectory with temporal and semantic transitions.

**Algorithm 1** Transition Probability Attack for Joint Time Period

**Input:** Anonymous set $S = \{x_1, x_2, \ldots, x_k\}$, time $t$, threshold $\theta$, former location $x$
**Output:** Number of identified positions *num*
$num \leftarrow 0$
**for** $i = 1$ to $k$ **do**
  $probability \leftarrow$ transition probability$(x, x_i)$
  **if** $x_i \in S_t$ and $probability < \theta$ **then**
    $num \leftarrow num + 1$
**return** *num*

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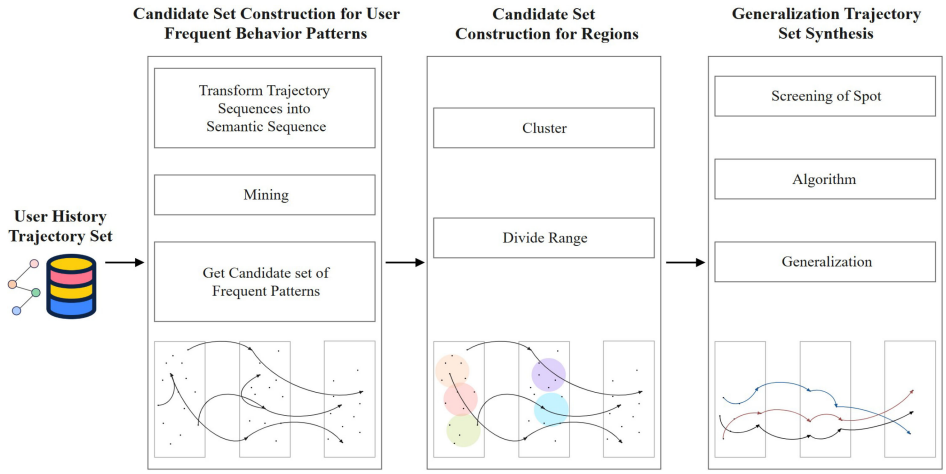



Fig. 2. Framework of DTS.

is used to categorize transition probabilities into three levels: 1) low: $\theta = 0.3$; 2) middle: $\theta = 0.4$; and 3) high: $\theta = 0.5$.

### C. System Model

As illustrated in Fig. 2, the system model comprises three components: 1) candidate set construction for user frequent behavior patterns; 2) candidate set construction for regions; and 3) generalization trajectory set synthesis. The user history trajectory set contains information, such as the location point, time, and longitude and latitude coordinates visited by the user. Making full use of this information is helpful in selecting the best candidate location points.

In the first component, user modeling, we address two primary challenges. The first challenge is managing the large volume of trajectory data while ensuring scalability. To overcome this, we partition the user's daily trajectory into multiple nonoverlapping time intervals, reducing the data volume per interval and enabling parallel processing for improved efficiency. This temporal partitioning also facilitates the synthesis process by ensuring it remains scalable across large datasets.

The second challenge lies in maintaining consistency with actual user behaviors during trajectory synthesis. To tackle this, we mine frequent behavior patterns from the trajectory data to capture typical user habits. These frequent patterns are then integrated into the synthesis process, ensuring that the generated trajectories reflect realistic user behaviors while preserving spatial relationships.

For the region candidate set construction, we apply density-based clustering methods, such as DBSCAN, to group location points within each time interval into regions of interest. This clustering process ensures that the candidate regions accurately reflect the user's movement patterns during different time periods.

Finally, in the trajectory synthesis phase, we use the region candidate set and frequent behavior patterns to generate generalized virtual trajectories. These synthesized trajectories are statistically similar to real user behaviors, making it difficult for attackers to distinguish between synthetic and actual trajectories, thereby ensuring privacy protection.

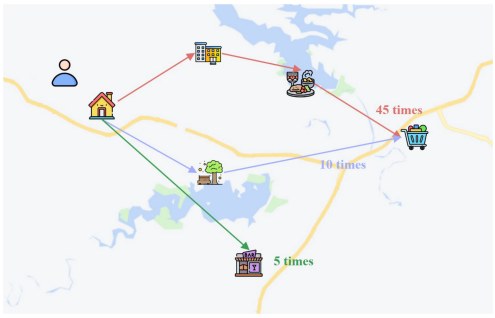



Fig. 3. Based on the user's trajectory data, we can conclude the following: the most frequent trajectory sequence for the user is "Home → Office → Restaurant → Supermarket," as it appears 45 times. The sequence "Home → Park → Supermarket" appears ten times, and "Home → Bar" appears five times.

### D. User Modeling

In the context of our methodology, the function module responsible for User Modeling plays a pivotal role in extracting valuable insights from user trajectory datasets. This module employs sequence pattern mining techniques from the field of data mining to identify and extract frequent sequences that meet the minimum support threshold. The process can be outlined as follows.

*Trajectory Data Preprocessing*: Prior to initiating the data mining process, it is imperative to preprocess the raw user trajectory datasets. This involves the transformation of trajectory sequences into semantically meaningful sequences, allowing for a more effective mining process.

*Sequence Pattern Mining:* The heart of our approach lies in the sequence pattern mining phase. We commence with mining sequence patterns from the prefix of length 1, and for each such prefix, we retrieve the corresponding frequent sequence by querying the associated projection database. This iterative process continues until it is no longer feasible to extend the prefix.

*Candidate Set of Frequent Patterns:* By the conclusion of the process, we have successfully derived the user's frequent sequence patterns, effectively creating a candidate set of frequent patterns that encapsulate significant user behavior patterns as shown in Fig. 3.

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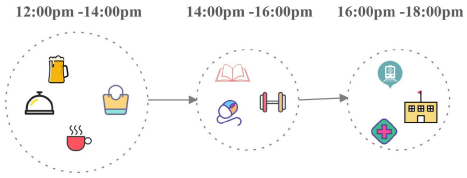



Fig. 4. Trajectory location points from different time periods are divided into different clusters as follows. The 12:00–14:00 time period includes Western restaurants, pubs, cafes, and shopping malls. The 14:00–16:00 time period includes libraries, Internet cafes, and gyms. The 16:00–18:00 time period includes stations, hospitals, and schools.

**Algorithm 2** Generate Candidate Area Set

**Input:** *user's track set* $T_i = t_1, t_2, ..., t_m$,*MinPts*,*eps*,*ts*
**Output:** *candidate area set* $A_i$

$A_i = \emptyset, \Omega = \emptyset, \Psi = \emptyset$
**for** $i = 1, 2, ..., m$ **do**
  **for** $j$ *in* $t_i$.*length* **do**
    $div(t_i, \Psi, ts)$
**for** $i = 1, 2, ..., m$ **do**
  **if** $|N_{eps}(\Psi_{p_i})| \geq MinPts$ **then**
    $\Omega = \Omega \cup \Psi_{p_i}$
$\Gamma = \Psi$
**while** $\Omega \neq \emptyset$ **do**
  $\Gamma_{old} = \Gamma$
  $Q =< o >$
  **while** $Q \neq \emptyset$ **do**
    **if** $|N_{eps}(q)| \geq MinPts$ **then**
      $\delta = N_{eps}(q) \cap \Psi$
      $\Psi = \Psi / \delta$
  $A_i = \Psi_{old} / \Psi$
  $\Omega = \Omega / A_i$
**return** $A_i$

In essence, the User Modeling module is a crucial step in our methodology, providing a comprehensive framework for mining and leveraging frequent user behavior patterns from trajectory data. This in-depth understanding of user behavior is instrumental in the subsequent phases of our research, ultimately leading to enhanced outcomes in our privacy-preserving trajectory synthesis approach.

### *E. Build the Region Candidate Set*

The main algorithm of the function module is Generate candidate area set algorithm, the main idea of Algorithm 2 is to get the candidate set (such as Fig. 4) of each time period by clustering the location points on the trajectory data set in each time period [40], [41]. The overall process of the algorithm is as follows.

1) The user's historical trajectory set is denoted as $\Omega$, and $tra_m$ is the mobile trajectory for a specific day. The parameter MinPts represents the threshold for defining core points, while *eps* is the neighborhood radius used for density definition. The set of time periods is denoted as *ts*, and the regional candidate set for each time period is stored in $A_i$. Specifically, the full-day timeline is divided into seven segments: 4:00–6:00, 6:00–8:00, 8:00–10:00, 10:00–12:00, 12:00–14:00, 14:00–16:00, 16:00–18:00, 18:00–20:00, 20:00–22:00, and 22:00–4:00 of the next day. This segmentation reflects typical human activity patterns across the day. The nighttime period (22:00–4:00) is deliberately extended due to the relative sparsity and higher individual variation of trajectories during these hours.
   For example, consider a user's trajectory for a day, where location points are recorded throughout the day. MinPts could be set to 5, meaning that a location point must have at least five other points in its neighborhood to be considered a core point. The *eps* parameter might be set to 10 m, defining the radius for neighboring points. These parameters ensure that only sufficiently dense clusters of points are identified as significant.
2) The position points for each trajectory in $tra_m$ are processed and divided into the corresponding time sets according to *ts*. The function *div* extracts the position points from the trajectory $tra_i$, matches each point to its corresponding time period in *ts*, and places the points into the respective time period set.
   For example, if the time period set *ts* contains early (6:00–8:00), afternoon (16:00–18:00), and evening (18:00–20:00), each location point will be assigned to the appropriate time period based on its timestamp.
3) The location points within the same time period are processed. For each location point, if the number of location points in a circle with this point as the center and *eps* as the radius, denoted as *n*, is greater than or equal to MinPts, this point is added to the core object set; otherwise, the next location point is considered until all points are processed.
   For instance, consider a location point $loc_1$ within a time period. If the number of points in its *eps*-radius neighborhood is 6 (and MinPts is set to 5), then $loc_1$ will be considered a core point and added to the core object set $\Omega$.
4) If the number of points in the neighborhood of location point $loc_i$ exceeds *MinPts*, the point $loc_i$ is added to the core object set $\Omega$.
5) Initialize the set of unreachable location points, denoted as $\Gamma$.
   For example, if a location point $loc_4$ does not meet the MinPts threshold in its neighborhood, it is added to the set of unreachable points $\Gamma$.
6) Randomly select a core object and initialize the queue $Q$.
7) Process the queue $Q$: if the number of other location points in the neighborhood of location point $loc_i$, denoted as $N_{eps}$, is greater than or equal to MinPts, add the point to the queue $Q$.
   For example, consider a selected core point $loc_1$. If the neighbors of $loc_1$ meet the MinPts condition, they will be added to the queue $Q$, and further expansion of the neighborhood will occur.
8) Finally, the regional candidate set $A_i$ for the time period is obtained.

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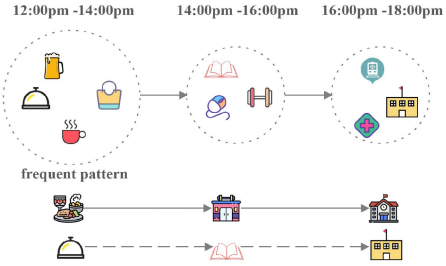



Fig. 5. Below the frequent trajectory is one of the false trajectories. Select a Western restaurant from the 12:00–14:00 cluster as the first false location in the trajectory, a library from the 14:00–16:00 cluster as the second false location, and a primary school from the 16:00–18:00 cluster as the third false location.

### F. Synthesize Generalization Trajectory

The main algorithm of the functional module is DTS algorithm. The main idea of Algorithm 3 is to first determine the transition relationship between regions according to the obtained candidate set of frequent patterns and the semantic information of each region. Then, based on the candidate set of frequent patterns and the position points in the region, the transition relationship between the position points is determined to form the generalized trajectory set (such as Fig. 5). In the function module, determining the location, before the transfer relationship between the need to distance formula to preliminary screening of a spot, the Manhattan distance [42] formula is

$$c = |x_1 - x_2| + |y_1 - y_2|. \tag{1}$$

The $(x_i, y_i)$ is the longitude and latitude coordinates of the location point.

The overall process of the algorithm is as follows.

1) The region candidate set $A_i$, the position point set P in the current period, the frequent pattern candidate set PS, and finally the generalization trajectory set $T$ are output.
2) First, the candidate region set is screened according to the location points, and the semantic similarity between the location points and regions is calculated by the Computes algorithm, where we adopt the Jaccard similarity to measure the overlap between their semantic tag sets. The top three regions with the highest combined semantic and spatial similarity scores are selected.
3) The second for loop is to synthesize the generalization trajectory. First, the region trajectory is synthesized, and the new region candidate set *A'* is matched with the user's frequent pattern, and the region trajectory is finally obtained. Then the location points in the region trajectory are matched with the frequent pattern again, and the generalization trajectory set $T$ is finally formed.

**Algorithm 3** DTS

**Input:** *the candidate area sets*: $A_i = \{a_1, a_2, ..., a_m\}$, *current time period position set*: $T_j^{u_i} = \{p_1^{u_i}, p_2^{u_i}, ..., p_n^{u_i}\}$,$k$,$n$,$ps[m]$

**Output:** $k-1$ *dummy trajectories*: $T_j^{u_i}(dummy) = \{t_1^{u_i}, t_2^{u_i}, ..., t_{k-1}^{u_i}\}$

```
temp = ∅
temp_tra = ∅
for l = 1, 2, ..., n do
    A'_i = computes(A_i, p_l^{u_i})
    Min[3] = ∅
    for j in A'_i do
        if p_j^{u_i} ≤ Min[0] then
            Min[0] = p_j^{u_i}
            break
        if p_j^{u_i} ≤ Min[1] then
            Min[1] = p_j^{u_i}
            break
        if p_j^{u_i} ≤ Min[2] then
            Min[2] = p_j^{u_i}
            break
    adds p_1^{u_i} into temporary set temp[l]
for l = 1, 2, ..., n do
    while temp[l] ≠ ∅ do
        temp_tra[l] = Match(temp[l], ps)
    adds p_1^{u_i} into dummy trajectories T_j^{u_i}(dummy)
return T_j^{u_i}(dummy) = {t_1^{u_i}, t_2^{u_i}, ..., t_{k-1}^{u_i}}
```

## IV. Performance Analysis

In this section, we present the results of a comprehensive set of experiments conducted to assess the performance of our protect scheme. We begin by outlining the experimental setup, which includes the hardware and software environment. We then discuss the datasets used, comparative methodologies, and the evaluation metrics employed. Subsequently, we report and analyze the outcomes of these experiments.

### A. Experimental Platforms and Datasets

The hardware and software environment for the experiment includes Windows 10 operating system, Core I5-6300HqCPU, and 16-GB memory capacity. This article studies the Gowalla dataset, which records the check-in records of users in the United States at various locations, and links the access points of users in a day into a movement trajectory according to the time. The locus contains information, such as the location point, the semantic relationship of the location point, and the time of check-in. In order to facilitate data observation and processing, the observation time window is set to one day, that is, 00:00–23:59. In the experiments evaluating privacy protection utility, we selected trajectories from 100 different users and conducted 12 experiments, each with anonymity set sizes ($K$ values) of 5, 10, 15, and 20, presenting the average results for each scenario.

### B. Evaluation Metrics

This study employs recognition rate and transfer direction deviation (DD) as evaluation metrics for privacy protection and service utility, respectively.

*Recognition Rate:*

$$\eta = \frac{\text{Recognized Locations}}{\text{Total Locations}}. \tag{2}$$

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The recognition rate evaluates the effectiveness of privacy protection against semantic attacks. A smaller set of identified trajectories indicates stronger privacy protection. The privacy protection strength is evaluated using (2). "Recognized Locations" refers to the set of trajectory points that an attacker successfully identifies via semantic attacks, which are considered as privacy leaks, thus introducing potential privacy risks. "Total Locations" refers to the entire set of possible location points in the user's trajectory dataset, including both identified and unidentified locations. The metric $\eta$ represents the probability of being recognized, and smaller values indicate stronger privacy protection.

*DD (Transfer Direction Deviation):*

$$\cos\theta^i \angle abc. \tag{3}$$

Here, $\theta^i$ represents the angle formed by the synthetic location $a$, the original location $c$, and the previous location $b$. The average cosine value of the deviation angles of all points in the anonymity set is used as the evaluation metric. A higher value of this metric indicates smaller deviation angles, suggesting better data utility and stronger resistance to velocity mutation attacks.

### C. Baseline

1) *DLS [13]:* The DLS method generates a set of fake locations with the highest query entropy based on the entropy value of geographic query probability and geographic query probability.
2) *RDG [23]:* The RDG method combines static geographic block query probabilities with dynamic transition probabilities to generate anonymous sets that enhance geographical continuity and query relevance.
3) *MSP [24]:* The MSP method constructs anonymous sets using the user's deep semantic tree, accounting for semantic relationships between users and locations to improve the quality and relevance of the sets.
4) *BSPri [11]:* The BSPri is a privacy-preserving method that annotates user stay points with semantic information, clusters them using a semantic tree, and generates synthetic trajectories via a hidden Markov model (HMM) while ensuring limited privacy gain for attackers through $\delta$-Privacy.

### D. Privacy Protection Strength Analysis

To assess the privacy protection effectiveness of various methods, we compare the performance of three different levels of the *transition probability attack for joint time period*. As shown in Fig. 6($\theta$=0.3), under low-intensity attacks, the *DTS method* consistently maintains a low recognition rate, demonstrating its robust privacy protection capabilities. As the parameter $K$ increases, which represents the number of synthesized dummy trajectories, the recognition rate remains stable and consistently stays below 0.2. This suggests that the DTS method is highly resistant to attacks, even when the number of dummy trajectories increases, preserving the privacy of users in LBSN. This superior performance can be attributed to the DTS method's fine-grained consideration of both semantic similarity and temporal alignment. By incorporating time-aware clustering and trajectory direction filtering, DTS ensures that the selected dummy points not only semantically resemble real locations but also fit the user's typical temporal behavior patterns. This dual-level filtering significantly reduces the likelihood that adversaries can distinguish real trajectories from synthesized ones.

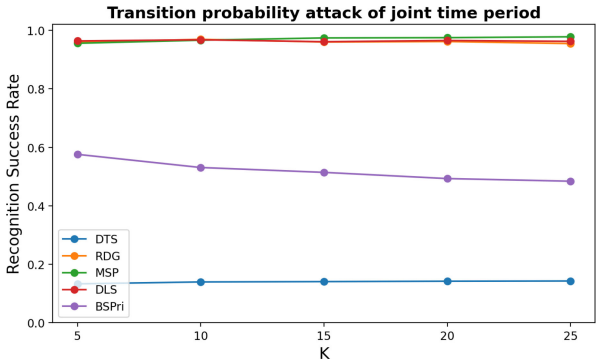



Fig. 6. Probability of different algorithms being recognized in the face of low-intensity attack [24].

In contrast, the *DLS*, *RDG*, and *MSP* methods show significantly higher recognition rates, above 0.9, under low-intensity attacks. While these methods experience a slight decrease in recognition rate as $K$ increases, their values remain exceptionally high, indicating vulnerability. These methods often construct dummy trajectories solely based on semantic transition matrices, which lack discrimination in temporal behavior. For instance, transitions between "home" and "work" may seem plausible semantically, but without temporal constraints, generating such transitions at unusual times (e.g., midnight) makes them easier to identify as fake. This temporal incoherence becomes a key vulnerability under joint time period attacks. As a result, these methods are unable to effectively counter the *transition probability attack of joint time period*, which leverages both semantic and temporal information to infer user behaviors.

The method BSPri performs better than DLS, RDG, and MSP under low-intensity attacks, maintaining a recognition rate between 0.5 and 0.6. While still higher than the DTS method, BSPri offers a notable improvement over the aforementioned methods, suggesting better privacy protection. This improvement can be attributed to BSPri's incorporation of both temporal and semantic factors, which allows it to better account for the context in which user behaviors occur, making it more resistant to attacks compared to methods that rely solely on semantic transition probabilities.

As the parameter $K$ increases, the recognition rate for BSPri gradually decreases, showing an improvement in attack resistance as the number of dummy trajectories increases. However, even with higher $K$ values, BSPri's recognition rate remains higher than that of the DTS method, indicating that DTS still provides superior privacy protection. This is likely due to DTS's more advanced handling of both temporal and semantic dynamics, further reinforcing its resilience against the *transition probability attack of joint time period*.

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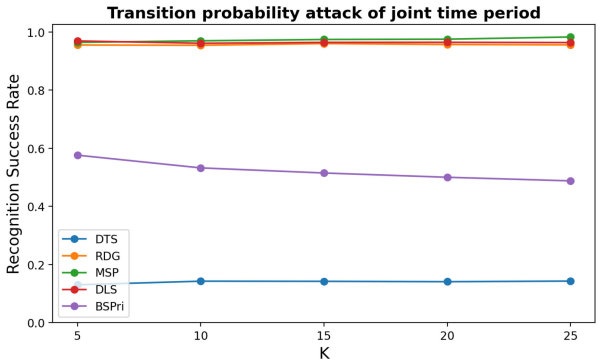



Fig. 7. Probability of different algorithms being recognized in the face of middle-intensity attack [24].

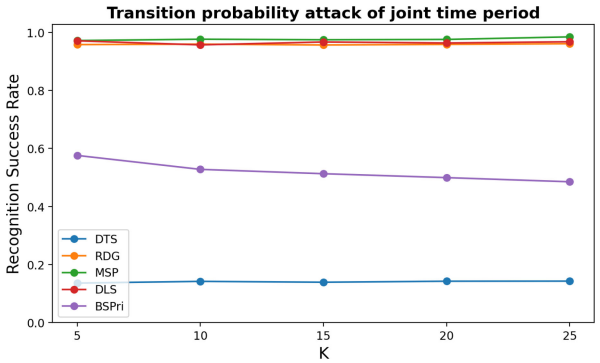



Fig. 8. Probability of different algorithms being recognized in the face of high-intensity attack [24].

As the attack intensity increases, as shown in Fig. 7($\theta$=0.4) and Fig. 8($\theta$=0.5) [24], the recognition rates for all methods increase, but the *DTS method* continues to demonstrate exceptional performance. Even under high-intensity attacks, the recognition rate for DTS remains below 0.2, a clear indicator of its resilience. Other baseline methods, such as DLS, RDG, and MSP, also experience an increase in recognition rate, which reflects their relative weakness under stronger attack conditions. While the increase is modest, it highlights the limitations of these methods, as their recognition rates were already quite high under low-intensity attacks. In comparison, the BSPri method shows a relatively stable recognition rate as the attack intensity rises, maintaining a performance level between 0.5 and 0.6. Its performance remains consistently better than DLS, RDG, and MSP, but BSPri's recognition rate is still significantly higher than that of the DTS method, which continues to outperform it under all levels of attack intensity. The results further emphasize the strength of the DTS method in providing robust privacy protection, even when subjected to increasing attack intensity.

### E. Service Quality Analysis

The *DD metric* [24], which measures the deviation of trajectory transition angles, is used to evaluate the impact of privacy protection methods on service quality. As shown in Fig. 9, the DD value for the *DTS method* does not fluctuate significantly with changes in $K$, indicating that the quality of service remains relatively stable regardless of the number of dummy trajectories generated. The *MSP method* exhibits similar behavior, with minimal fluctuation in DD values across different $K$ values.

In contrast, the *DLS* and *RDG* methods experience noticeable decreases in their DD values as $K$ increases. This decline suggests that these methods fail to adequately account for the impact of transition angles on service quality. The *DTS* and *MSP* methods, on the other hand, consider the influence of transition angles when generating location points, ensuring that the synthetic trajectories maintain a high level of spatial coherence and relevance. This is particularly important in LBSN, where maintaining the quality of service while ensuring user privacy is critical. The DD metric of BSPri increases with

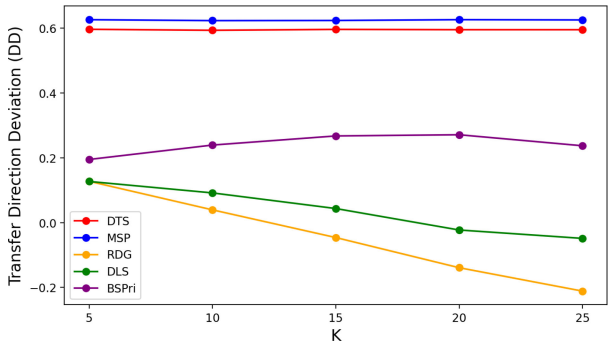



Fig. 9. DD metric [24].

the value of $K$, but begins to decrease once $K$ reaches 20, indicating its insufficient consideration of transition angles.

When comparing the DD values, *MSP* achieves the highest DD value, indicating superior service quality. The *DTS method* follows closely behind, with only a slight difference in the DD values. This minimal difference suggests that DTS achieves a good balance between privacy protection and service quality, ensuring that the generated dummy trajectories still maintain realistic transition patterns. The DD values of BSPri consistently remain between 0.2 and 0.3, which is notably lower than those of the *MSP* and *DLS* methods. On the other hand, the DD values for *DLS* and *RDG* are both lower than 0.2, indicating a significant decline in service quality. Notably, as $K$ increases, the *RDG method* even shows negative DD values, which correspond to transition angles greater than 180°, signaling a substantial deviation from realistic trajectory patterns. This highlights a critical limitation in the RDG approach, where the lack of consideration for transition angles results in poor service quality.

### F. Overall Evaluation

In summary, the *DTS method* exhibits outstanding performance in both privacy protection and service quality across various attack intensities. In the privacy protection experiments, DTS consistently maintains a low recognition rate, demonstrating its robustness against different levels of the *transition probability attack for joint time period*. Although the DD metric for DTS is slightly lower than that of the *MSP*

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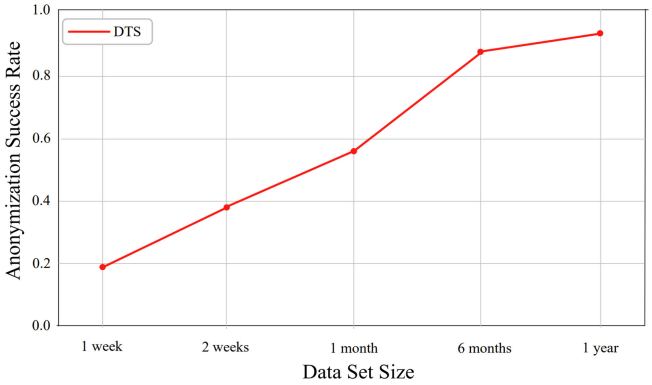



Fig. 10. Impact of dataset size on anonymity success rate.

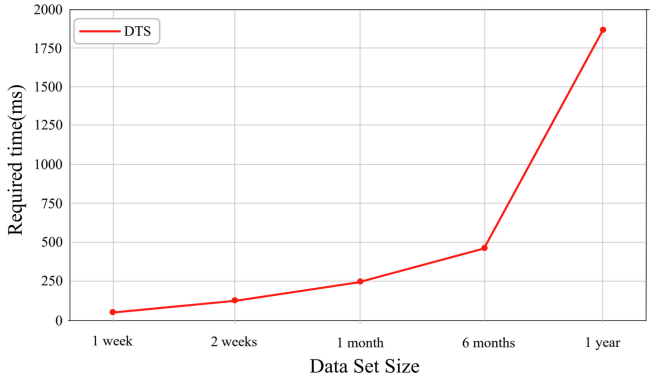



Fig. 11. Impact of dataset size on execution efficiency [43].

*method*, it still performs well, maintaining good service quality while providing strong privacy guarantees.

Overall, the *DTS method* stands out as a superior solution for balancing privacy protection with service quality. It outperforms all baseline methods, excelling in maintaining both data utility and user privacy, and demonstrating resilience against various attack strategies. The results show that DTS offers a promising approach for ensuring privacy in LSBNs, making it an effective tool for privacy-preserving trajectory data publishing.

### G. Anonymous Success Rate Analysis

This experiment considers the influence of the size of the dataset on the success rate of anonymity of the model. Because this experiment anonymizes the user's track from the user's behavior and habits, theoretically, if the more historical data of the user is more detailed, the user's behavior and habits model obtained after modeling the user can more accurately simulate the real behavior of the user. As can be seen from experimental results shown in Fig. 10 [24], the success rate of anonymity increases with the size of the dataset gradually increasing, and the success rate increases significantly when the dataset reaches a certain size, which indicates the overall modeling.

In addition to evaluating the success rate, it is also important to consider the time complexity of the anonymization process. Specifically, the time complexity for the online anonymization set release is $O(n \cdot m^2)$, where $n$ represents the number of user trajectory data points, and $m$ is the size of the candidate region set. While larger datasets lead to improved anonymity success rates, the time complexity also increases with the dataset size, which may impact execution efficiency. Therefore, balancing privacy protection and execution efficiency is crucial, particularly in real-time applications where performance is key.

### H. Anonymous Efficiency Analysis

As shown in Fig. 11, the processing time [43] of the proposed scheme is less than 250 ms for datasets smaller than one month. When the dataset size reaches six months, the processing time is around 500 ms. For datasets of one year, the processing time exceeds 1500 ms. The time overhead for anonymization increases significantly with the dataset size, especially from six months to one year.

To achieve better anonymization results, Fig. 10 indicates that larger datasets lead to higher success rates. However, Fig. 11 shows that larger datasets also result in increased overhead. Additionally, the improvement in success rate from six months to one year is similar to that from three months to six months, but the time overhead increases sharply. To accurately model user behavior while maintaining high execution efficiency, using six months of user trajectory data with the DTS method is recommended for optimal performance.

### I. Engineering Applications

The DTS method offers significant applications in various engineering domains, particularly in smart city infrastructure, transportation systems, and mobility services, where privacy concerns are critical. In smart cities and urban mobility, DTS enables the generation of anonymized mobility patterns for traffic management and urban planning. It allows city authorities to optimize traffic flow and public transportation schedules without compromising user privacy. For ride-hailing and transportation services, DTS protects user location data while still enabling optimal route planning and demand prediction. By synthesizing anonymized trajectories, these services can ensure privacy while maintaining operational efficiency. Additionally, in location-based services and marketing, DTS can be applied to anonymize user movement data, ensuring privacy while still allowing businesses to deliver location-based advertisements and personalized services. In autonomous vehicles and traffic management, DTS ensures privacy preservation when collecting data to optimize traffic systems and enhance safety, without exposing individual users' location histories. These applications demonstrate the versatility of the DTS method in protecting privacy across various real-world systems, all while maintaining the utility of the data for analysis and optimization.

## V. Conclusion and Future Work

In this article, we proposed the DTS method, an innovative privacy preservation solution for LBSN. DTS improves user privacy by synthesizing generalized trajectories that align with actual user behavior. By employing data mining techniques, we

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model user behavior patterns and divide time into nonoverlapping intervals for efficient trajectory clustering and synthesis. Experimental results on real-world datasets show that DTS outperforms existing methods in terms of privacy protection and service quality.

While DTS significantly enhances privacy protection, it has some limitations. First, synthesized trajectories may not always align perfectly with road networks, potentially impacting privacy performance. Additionally, in urban environments with dense and uneven distributions of points of interest (POI), the effectiveness of DTS could be compromised, especially in grid-based partitioning schemes.

For future work, we aim to address these limitations by incorporating road network factors into the privacy protection mechanisms and exploring semantic trees to handle dense POI distributions. Furthermore, we plan to enhance feature capture in privacy schemes by leveraging the frequent user pattern mining approach, improving privacy protection in challenging environments. These improvements will further optimize the DTS method for practical deployment in diverse LBSN applications.

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